AUtoMLy Medicaid Policy Copilot

Milestone 1 — Data & Model Selection: Post-Feedback Revision

J.O. • INAI 6986 AI Capstone • Evidence measured through October 8, 2026 (Central Time)

Reporting Boundary

This post-feedback revision uses measured development evidence through October 8, 2026. AUtoMLy is the policy-copilot project; ALPHIN is its adapted-model component. The original submission is preserved. Nemotron is the student-accepted candidate proposal, not a deployment-ready or final-frozen system. No protected final evaluation has run. Supporting records follow the core narrative.

1. Data Characteristics

The prepared corpus is a frozen snapshot (September 14, 2026) of 138 public Alabama Medicaid PDFs (64 Administrative Code files, 74 Provider Manual files; 2,433 pages). Preparation produced 1,999 ordinary question–answer records with source URL, page, exact quote, and source SHA-256 for every row; one blank-answer safety case was quarantined, and the retired September 16 generator output (326 records) was withdrawn to procedural negative evidence only. Preparation is reproducible in code and documented by the frozen manifests and preparation code. Current result replay is a separate artifact.

Historical preparation split: grouped by source document, assigned by SHA-256 hash bucket, so an entire document lands on one side of the boundary. Frozen historical counts: train 1,684 records / 113 documents; validation 124 / 10; historical test 191 / 13; plus 30 separate student-authored control records. The current development study adds a nested token-fit design: seeded document round-robin subsets of 200, 509, 750, and 1,600 training rows, with every subset nested and the smallest covering every training document.

What changed since the Week 4 gate and the original submission: the instructor's split-exposure question was answered directly (Section 6); sixteen cross-split duplicate-answer groups covering 57 records were human-dispositioned with one near-duplicate quarantined; a four-family development shortlist on 24 fixed questions was run base-versus-adapter under matched conditions; and a candidate proposal was accepted by the student (Section 5). Target definition is unchanged: supported-answer generation with a source locator, and abstention when sources do not support an answer.

This is supervised text generation, not a classifier with a feature-importance ranking. Inputs are policy questions; targets are sourced answer text and locator formatting. Preparation retains the question, answer, source passage, document identity and page, then renders the pinned native conversation format. Those fields connect each answer to evidence and make formatting effects visible. The ordinary corpus and separate quarantined safety case do not provide a representative unanswerable distribution; abstention coverage requires a dedicated audited stratum.

2. Candidates Considered

Following the feedback that two same-recipe Muse renders were a sensitivity run rather than a real candidate comparison, the comparison was rebuilt around capacity and configuration across current open-weights families that fit the single-GPU local environment. All four candidates train on the same nested subset policy (N=1,600 adapter selected) and serve locally; none is treated as a factual winner at this stage.

Candidate (adapter on base)	Checkpoint	Why plausible for this data and decision
Nemotron 3.5 Lightning 30B-A3B	checkpoint-400	Local base-plus-adapter inference completed every development slot. Its unchanged checkpoint also has completed human error-review and targeted-recheck evidence.
Gemma 4 26B-A4B-it	checkpoint-400	Different instruction-tuned family with a smaller active parameter count and completed nested-size training; its native template and adapter integration were verified before comparison.
Ornith 1.5 35B-A3B	checkpoint-400	Current-generation alternative of similar active size; native answer channel and adapter toggle verified before the campaign.
Muse Glimmer 30B (4-bit)	checkpoint-200	The original Milestone 1 family (custom template and answer-masking format, ATEM); admitted with its native high-reasoning setting unmatched because it has no supported reasoning-off condition — a disclosed comparability limit, not an exclusion.

The original candidates — two same-base Muse QLoRA renders (1,684 vs 1,685 rows, 211 steps) — are retained exactly as the instructor characterized them: a sensitivity run, not an independent candidate. The retired September 16 run remains procedural historical evidence only; its numbers are not mixed into any active comparison.

3. Baseline

Two baselines carry forward from the original submission, plus one new matched baseline. (a) Operational baseline: manual document search and drafting by the student on the 30 student-authored control questions — 30/30 source-validated; the feedback reports about 84 seconds per question, a retrospective session estimate rather than a controlled stopwatch measurement. As the feedback states, 100% on exact-string lookups is a ceiling, and the baseline therefore cannot discriminate on those items; it is reported with that limit, and no new controlled timed baseline has been run. (b) Technical baseline: the untouched base model with the same prompts, deterministic decoding, and token budget as the adapter. (c) New for this revision: each of the four candidates' untouched base is scored on the matched 24-question development set under identical runtime conditions as its adapter (Section 4), which is the baseline structure the feedback asked for.

Development scoring change, explicitly separated from the Project Management Plan: Answered AND Correct AND Supported AND no unsupported claims, per 24 slots per condition, where every slot — including budget failures and parse failures — stays in the denominator and abstention is counted separately. This post-hoc local-advisory v2 definition differs from the previously committed primary and is machine-advisory only. The plan committed to at least a 10-percentage-point answerable factual-accuracy gain over the untouched base, at least 95% valid locators, under 2% unsupported claims, and at least 90% correct abstention. The stricter composite is an additional support-quality measure, not a retroactive replacement for factual accuracy. Changing the baseline family from Muse to Nemotron follows the student-accepted candidate decision; final metric denominators and thresholds must still be frozen.

4. Evidence

Development shortlist: 24 fixed predeclared questions, base and adapter per family, matched same-question full-budget runtime, 192 slots total. Local advisory judge (post-hoc v2 correction), not human scoring. All 192 slots replay-verified: 150 accepted judgments, 19 budget-truncation failures, 23 parse failures, 0 unresolved, 0 retries; 17 answer successes and 48 abstentions across all conditions. No tuning was performed on these outputs.

Family (24 questions per condition)	Base successful answers	Adapter successful answers	Adapter budget failures	Adapter parse failures	Adapter known-unsupported (of judged)
Nemotron 30B-A3B	4/24	4/24	0	0	19 of 24
Gemma 26B-A4B	0/24	4/24	1	0	18 of 23
Ornith 35B-A3B	1/24	4/24	1	0	18 of 23
Muse Glimmer 30B	0/24 (16 budget)	0/24	1	23	not judgeable (parse failures)

Reading limits, stated plainly: the 4/24 adapter results for Nemotron, Gemma, and Ornith tie; small post-hoc machine-rated samples do not certify a winner or deployment accuracy. Muse's 0/24 is mostly a format/usability finding from its conservative answer-channel parser plus its unmatched native high-reasoning condition, not 24 incorrect policy judgments. The v1 protocol that had scored Ornith base 24/24 counted blank refusals as successes; that v1 result was withdrawn as a scoring correction — the outputs themselves were not rerun.

Historical results are retained only as an exposure and decision record. The original simple-lookup control ceiling and URL-presence gains do not establish policy-answer quality. September 16–17 superseded model runs are excluded from current comparison tables. The historical 191-row test was already exposed and cannot serve as a newly untouched final benchmark. The 16 current nested-size training runs measure within-family next-token fit, not cross-family factual superiority.

5. Selection & Justification

Selected candidate (proposal state): Nemotron 3.5 Lightning 30B-A3B, adapter sha256 4aaca5c5…c0a65a (checkpoint-400, train size 1,600, revision a9904d24…), evaluated only in a pair with its matched untouched base under identical conditions. The student accepted this proposal on October 8 under an explicit benchmark-scope decision; this is a user-accepted proposal, not a final system freeze, and the final run is not approved.

Nemotron ties Gemma and Ornith at 4/24 successful adapter answers, so the score does not establish a winner. It is proposed on operational evidence: 48/48 usable generation completions; a pinned base/adapter lineage; and the same adapter used in the completed human review (30 cards, 15 paired questions, 180 ratings) and October 7 source-honesty recheck (8 pairs, 16 new outputs). That recheck did not produce a successful fix. Local execution, source-verifiable citations, and conservative handling of unsafe answers are business requirements. A merge, deployment benchmark, latency advantage, or model-size cost advantage is not claimed from these records. Known unsupported claims in 19 of 24 adapter judgments and invalid locators in 14 of 24 make human source review mandatory.

6. Interpretability & Leakage Check

The feedback's central leakage question is now answered with artifact evidence rather than assertion: all 30 student-authored control items were mapped to their source documents' split. Ten control-source documents are assigned to the historical generated test split; zero control-source documents overlap the train or validation splits. So the adapter was never fine-tuned on the control set's source pages — the control measures source retrieval, but it interrogates documents historically exposed through the test rows, and that exposure is disclosed rather than claimed away. The questions themselves are student-authored; the source documents are not independent of the historical test.

Duplicate-answer leakage: the sixteen cross-split duplicate-answer groups covering 57 records that the feedback required ahead of scoring were dispositioned by the student on October 5–6 (27-card review: 16 exact groups, one near-duplicate pair ND01 quarantined with H2011 table-column case, nine candidate unanswerable cards reviewed). These dispositions do not retroactively repair already-trained weights; their remaining lineage impact must be reported. Model-error interpretation uses the v2 advisory error categories on the Nemotron pair: base errors concentrate in invented-rule citations (11) and wrong values (6); the adapter shifts the

failure mass to wrong values (15) and invalid locators (14 of 24 judged). Its substantive output increases while locator quality degrades — a business-relevant finding for citation review, not a reassurance. The untuned generator-convention lesson (URL presence ≠ grounding) is carried forward everywhere these numbers appear. Unaddressed strata: systematic fresh-benchmark coverage of exceptions, effective dates, conflicts, synthesis and abstention remains unverified. Earlier diagnostic cases do not close that coverage requirement.

7. What Would Change My Mind

Before the final freeze, I would reconsider Nemotron if a separate authorized development comparison shows another candidate delivering more human-verified, supported answers within the same runtime budget, or if new development evidence shows unacceptable citation or abstention risk. After the winner-only final campaign, failure to meet the plan's accuracy, locator, unsupported-claim, or abstention thresholds means report the negative result and withhold promotion. It does not authorize testing rival candidates on the same protected cohort or tuning against it. Any later redesign must use a separately governed development cycle and a new protected evaluation set. The October 7 failed refinement already argues against assuming that a source-honesty instruction can repair missing knowledge.

8. Repository

The source-of-record repository is private; its instructor access still requires confirmation. The historical notebook link is verified below. The new replay notebook must reproduce the current development tables from saved per-item outputs before this revision is submission-ready.

Artifact	Location / status
Private repository (source of record)	https://github.com/porespellar/INAI6986 — private; instructor access confirmation remains a pre-submission gate. Not published or submitted by this document.
Historical deliverables repository (public)	https://github.com/porespellar/INAI6986_project_deliverables — public, verified October 8. It contains notebook/milestone1_reproducibility.ipynb. This is the historical notebook; no fresh-kernel verification of October 8 results is claimed from it. Private grading copies are not published there by this review.
New development replay notebook	reproducibility/development_replay.ipynb — being finalized under a separate task; referenced but not claimed as freshly executed at this build. No fresh-kernel pass is claimed here.

Private source-of-record repository | Historical Milestone 1 notebook

This revision's numbers recompute from the hashed JSON/JSONL artifacts listed in Appendix B rather than from chat context; the builder asserts each count against its artifact at build time.

AI Use Disclosure

Hermes Agent (GPT-Sol 6.1) reviewed and edited this report; local Qwen agents assisted coding and drafting. J.O. conceived the project, chose its architecture and hardware, directed and troubleshot local runs, reviewed evidence and made proceed/quarantine decisions; he retains final accountability. The original controls are student-authored. Any new benchmark questions will be locally AI-generated and source-checked, not represented as independently student-authored.

Appendix A — Evidence Limits, Corrections, and Open Gates

Item	Exact status (no waiver is implied)
Advisory v2 scoring	Post-hoc correction to the original scorer on unchanged outputs (v1 Ornith-base 24/24 refusal inflation withdrawn). Local machine advisory only — not human adjudication, not final accuracy, not a rubric waiver, and not the previously committed primary metric presented as unchanged.
Matched denominators	24 slots per condition retained in every denominator; budget/parse failures count as not-correct; abstentions reported separately (48 total).
Muse comparability	Native high-reasoning setting unmatched (no supported reasoning-off condition); adapter 23 parse failures / 1 budget failure; base 16 budget failures and 8 judged abstentions. Disclosed comparability limit; not an exclusion without explanation.
Nested token-fit runs	Four families, 16 runs × sizes 200/509/750/1,600; cross-tokenizer loss is fit evidence only, never a factual ranking.
September 16–17 runs	Superseded; excluded from active comparisons; retained as procedural historical negative evidence; numbers not mixed.
Original 30 student questions	Complete and source-verified; all 30 map to historical test-source documents, none to train/validation. Historically exposed; no untouched claim. Student has explicitly declined authoring further questions; the proposed fresh benchmark will be locally AI-authored and source-checked with that authorship disclosed — an authorship-method change, not an instructor waiver.
Feedback coverage ask (reasoning / exception / date / conflict / 8–10 unanswerables)	Nine candidate unanswerable cards were reviewed October 5–6; their question wording is student-adopted/edited AI-assisted local-LLM draft (Batch A receipt c54e1ee5ef0e: 8 identical to the local-LLM draft, 1 student-edited) and is not represented as independently student-authored. They do not establish a completed fresh benchmark or its measured reasoning/exception/date/conflict/synthesis/abstention coverage. Earlier diagnostics and the fresh-source plan are not substitutes for that evidence.
Manual baseline	30/30 ceiling with retrospective ≈1:24/question estimate retained; no new controlled timed baseline; no measured time-savings claim.
Human Batch B	Submitted revision 11: 30 error-selected cards, 15 paired questions, 180 ratings. Scope: original error review — not the 192-slot shortlist, not population accuracy.
October 7 refinement	8 pairs / 16 outputs re-generated on the unchanged adapter (hash verified); 32 initial advisory labels with 5 schema failures distinct from v2, no retries. Negative finding on blank refusals / bad citations; human scores of refined outputs absent. Measures instruction sensitivity and observed citation errors, not source-context ablation, attention/saliency or feature importance; no final robustness delta.
Gates	Final system/key freeze: not approved. Final model campaign: not authorized. Human hours: not supplied, not fabricated. Protected final prompts/keys: not opened.

Appendix B — Evidence Ledger (relative to project root)

Claim(s)	Artifact (asserted at build time)
1,999 records; 1,684/124/191 splits; 113/10/13 documents; nested subset policy; 30 controls excluded; 10 control-source docs in historical test, 0 in train/validation	artifacts/model_comparison/2026-09-25/nested-train-subset-manifest.json
Matched 24-question shortlist contract, settings, prompt IDs, exposure statement	artifacts/milestone2_completion/development/shortlist_v1/frozen/contract.json
192-slot v2 advisory results; per-condition numerators/denominators; error categories	…/verification/parent_advisory_v2_actual_results_verified_20261008.json; development_comparison_corrected_v2_20261008.md; development_v2_counted_outcomes_20261008.jsonl
Campaign totals (150/19/23/0; Muse caveat; no winner; final_evaluation=false)	…/verification/four_family_collection_summary_20261008.json
Batch B rev 11 counts; refinement lineage; no final approval	…/verification/parent_development_gate_reconciliation_verified_20261008.json
Candidate proposal; fresh-source scope; authorship disclosures	…/verification/user_candidate_benchmark_scope_decision_20261008.json; class_package_20261008/evidence_snapshot.json

Claim(s)	Artifact (asserted at build time)
Batch A dispositions (16 groups/57 records + ND01/H2011 quarantine; 9 UNSUP cards reviewed; receipt scope)	artifacts/milestone2_completion/review/responses/submission.json; Batch_A_Receipt_c54e1ee5ef0e.json
Historical F1 / numeric / citation table; fail-closed lineage; manual-baseline pilot	artifacts/milestone1/2026-09-22/evaluation_snapshot.json; human_scoring/2026-09-24/human_generated_sample_scores.json
Duplicate audit provenance (16 groups / 57 records; LLM-review-only pilot)	artifacts/milestone1/quality_review/2026-09-24/sample_manifest.json
Full hash schema for all above (SHA-256 per file)	class_package_20261008/evidence_snapshot.json and parent_review/milestone1_personal_review.json; source hashes are recorded in the review manifest

Appendix C — Original 30 Student-Authored Control Items (Reference)

Per the feedback, the 30 single-record appendix pages are collapsed to one table. Full question text, expected answers, and quotes remain in the preserved original submission and the frozen control file (data/curated/student-authored-control-set-2026-09-22.jsonl, sha256 a1deb523… bea55c); they are not reprinted here to avoid re-exposing the answer key in a revision draft. Every item is student-authored, browser-verified, training-excluded at the record level, single-token code/page lookup in character, and maps to a historical-test-source document.

Source document and page	Items	Control IDs (order not sequential)	Question type (as criticized)	Source-split mapping
Jul26_34.pdf, page 2	10	SA-010–013, SA-021–026	Specialty-code table lookups	Historical test
Jul26_28.pdf, page 2	6	SA-008, SA-014, SA-015, SA-028–030	Provider type / code lookups	Historical test
Apr26_37.pdf, page 2	5	SA-006, SA-016–019	Provider type / code lookups	Historical test
Jul26_111.pdf p.1; Jul26_30.pdf p.1; Apr26_23.pdf p.1	5	SA-005, SA-007, SA-009, SA-020, SA-027	Code / page lookups	Historical test
Administrative Code single-page chapter files (Ch. 1, 28, 49, 50)	4	SA-001–004	Table-of-contents lookups	Historical test

The original lookup set remains concentrated: 21 of 30 items sit on page 2 of three manual files and four more are single-page table-of-contents lookups. That is 10 source documents in total, all in the historical test split. The concentration is the reason the fresh benchmark re-samples sources; the unanswerable and reasoning strata remain unscored until that benchmark is frozen (Appendix A).

End of Milestone 1 post-feedback revision draft. Awaiting personal rubric review and edits; no grade is claimed or implied.